

Supplementary material for "Constrained, not identifiable: how the Voynich manuscript text was produced"

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S1. Exploratory profile: materials and methods

Parsing. Page headers of the form `<flr> <! $Q=A $P=A $F=a $B=1 $I=T $L=A $H=1 ...>` supply quire, illustration section, Currier language and hand. Locus lines `<flr.1,@P0>` text carry a locus type whose first letter, after stripping the position markers `@ + * = ~ ! / &`, gives the kind: P (paragraph text), L (label), C (circular), R (radial). Loci whose type is prefixed by `&` (labels placed in drawings) are labels. Inline markup: alternatives `[a:o]` take the first reading; extended glyph codes `@nnn`; and `?` mark a token as uncertain, and such tokens are removed; comments `<! ... >` and annotations `{...}` are removed; drawing intrusions `<->` and `<~>` are spaces; paragraph markers `<%>` and `<$>` are removed; fillers `!` are deleted. A `.` is a space (35,750 occurrences) and a `,` an uncertain space (9,316), joined in the baseline. Ligature and capitalised variants are kept as distinct glyphs. Result: 4,130 paragraph-text lines on 207 pages, 32,428 tokens; 1,114 label tokens.

Line breaks. For consecutive lines of the same paragraph, the excess MI of the pair (last token of line k , first token of line $k+1$) is compared with that of adjacent within-line pairs subsampled to the same n (eight subsamples). For consecutive lines of the same page we also compute the excess MI of all combinations of position classes, and the page specificity $I(\text{token}; \text{page})$ by position in the line.

Boundary predictability. Spaces are removed from each line and a model $P(\text{boundary} \mid k \text{ glyphs on the left, } k \text{ on the right})$ is fitted on a random half of the lines and evaluated on the other half. We report the information gain $1 - H(\text{boundary} \mid \text{context})/H(\text{boundary})$ and F_1 at threshold 0.5. The rule-spaced Latin control is stripped of spaces and re-spaced after `m`, `s`, `t` or `y` once the current token has at least three letters.

Lexicon test. A within-token glyph model with two glyphs of context and token start and end symbols is fitted on a random half of the tokens, and tokens are sampled from it. We compare the share of held-out real tokens (and types) whose string occurs in the training half with the same share for the sampled tokens. An open inventory defined only by local glyph constraints gives equal shares; a lexicon gives a large gap.

Entropy profile and residual. Each line is a string with the space as a symbol. Plug-in conditional entropies $h(k) = H(x \mid k \text{ preceding symbols})$, $k = 1-5$, are compared with synthetic twins sampled from Markov models of order 1–3 fitted to the same corpus with the same line lengths; only the comparison with the twin is interpreted, because plug-in $h(k)$ is biased downward for large k . The residual is the lzma rate (preset 9, bits per symbol) of a text minus that of an order-3 twin fitted separately for each combination of Currier language and section. A negative residual means redundancy the drifting automaton does not capture. The automaton's own samples (about ± 0.03 bit) set the noise floor.

Generators. All generators produce text in the line-length, paragraph and page structure of the Voynich paragraph text.

- *Glyph automata.* Counts of order- k contexts over the glyph stream of each line with k start symbols and an end symbol; sampling per line with the target glyph length of the corresponding real line, the end symbol suppressed until 60% of the target length and sampling stopped at 140%. Section drift: one model per (Currier language, section) group

with at least 30 lines, otherwise the global model. Page drift: the next-symbol distribution is λ times the page's own normalised counts plus $(1 - \lambda)$ times the section's, for contexts seen on the page; this variant sees the page's own text and is in-sample by construction.

- *Table and grille* (after Rugg 2004). Slot decomposition of each attested token by the regular expression $\text{prefix} = q?(o|y|d|s|ch|sh)?(k|t|p|f|ck|ct|cp|cf|ch|sh|ckh|cth|cph|cfh|l|r|d|s)?$, $\text{midfix} = e\{0,3\}(o|a|y)?$, $\text{suffix} = (i\{0,3\}(n|r|l|m)|d|l|r|s|y|dy|ly|ry|dl|ol|al|or|ar|am|an)?y?$, with unmatched tokens split into thirds. Per section group a table of 40 columns and $\max(2, \text{round}(0.9 \times \text{tokens}/40))$ rows, each cell an independent draw of prefix, midfix and suffix from the empirical frequencies. Each page starts at a random cell; a line is the run of cells to the right (wrapping); each line advances one row.
- *Copy and modify* (after Timm and Schinner 2020). Seeds: the 300 most frequent real tokens. Source: line above, same column ± 1 (0.6); previous token in the line (0.2); random token of the page (0.15); seed (rest). Modification with probability 0.85, a second with 0.25: substitution within the similarity pairs (ch,sh), (k,t), (t,p), (k,f), (e,ee), (ee,eee), (i,ii), (ii,iii), (a,o), (y,dy), (r,l), (n,m), (ol,or), (ar,al), (ain,aiin), (ey,edy) with probability 0.5; otherwise addition or removal of a prefix from {q,o,d,s,y,l,ch,sh} (0.25) or a suffix from {y,dy,n,in,iin,r,l,s,m} (0.25), chosen so that tokens stay around five to six glyphs.
- *Verbose cipher with fixed homophone choice*. Each of 23 Latin letters maps to one to three strings from a pool of 35 Voynich-like units; the homophone is selected by the code of the previous glyph modulo the number of homophones. Spaces after y, n, r, l, m, s once the current token has three glyphs, forced at nine. One independent Latin sentence per line.
- *Ciphertext-feedback substitution*. $c_t = T[c_{t-1}][p_t]$ with 27 rows, each a random injective map from 23 letters to 27 glyph symbols; the two-glyph variant applies a second table with the intermediate glyph as state. Spaces after y, n, r, l, m at a minimum token length of three and a maximum of nine. Independent sentences per line.
- *Naibbe cipher* (Greshko 2025). Tables from the file `naibbe_tables.csv` of the author's repository (138 unigram, 138 prefix and 138 suffix strings for 23 letters and six tables). Preprocessing: lower case, $w \rightarrow uu$, $j \rightarrow i$, $k \rightarrow c$, letters only. The plaintext is cut into one- or two-letter tokens with unigram probability 17/36 (the last letter of a line is always a unigram). Deck: 20 alpha, 8 beta1, 8 beta2, 8 beta3, 4 gamma1, 4 gamma2, shuffled and dealt without replacement, reshuffled when exhausted; one card per plaintext letter (variant: per token). Two-letter tokens are $\text{prefix}(\text{first letter}) + \text{suffix}(\text{second letter})$. Each ciphertext space is removed with probability 0.03 (variant: none). Plaintext: Latin (Caesar) or Italian (Dante), one independent sentence per line (variant: running text). State-dependent variant: with probability w the table of the next token is drawn from a fixed random distribution over the six tables conditioned on the last glyph of the previous token (one Dirichlet(c) draw per glyph), otherwise from the deck.

Drift and hand. For pairs of pages of the same Currier language we compute the Jensen–Shannon divergence between their glyph-bigram distributions (450-glyph samples, additive smoothing 0.05) as a function of folio distance, grouped by whether the pages share section and hand. Hands follow Davis (2020) as recorded in the ZL metadata.

Illustration features and text–image test. Each page image of folios 1r–58v (free of fold-outs; folio 12 is missing) is rendered at 50 dpi from the Beinecke digital facsimile and cropped by 6% on each side. Pixels are classified relative to the page's median (parchment) colour into green, blue and red–brown. Features per page: the area shares of the three classes, their vertical and horizontal centroids, the mean hue of the green class, the blue and red shares of all coloured pixels, and the total coloured share (standardised). The page-to-folio assignment was verified visually on f26r, f27r and f58r. Text distances are Hellinger distances between relative token-frequency vectors (and, separately, glyph-bigram vectors); image distances are Euclidean. Their Spearman correlation over

page pairs is tested by permuting page identities of the image matrix (10,000 permutations), excluding pairs closer than three folios and, for page sets with several hands, permuting within hand. The positive control adds k pseudo-tokens per page that are deterministic functions of four binary features (blue share, green area, red share, vertical centroid). A secondary analysis predicts the section of a page within one hand from its token or bigram frequencies (leave-one-page-out nearest centroid, excluding folio neighbours within a radius), against the majority rate and the nearest remaining folio; it is confounded by folio position because sections occupy contiguous folio ranges.

Labels. For each illustrated section, the share of label tokens whose type occurs on at least one other page of the section is compared with that of equally many randomly drawn paragraph-text tokens of the same pages (or, where a page has too little text, of the same section), averaged over 20 draws.

Uncertainty. A page bootstrap with replacement was tried first and rejected: duplicated pages create identical lines and token pairs, which inflates plug-in MI (token MI interval 0.49–0.71 around a point estimate of 0.08) and depresses the hapax share. The leave-one-quire-out jackknife (16 quires) has no duplication and is used throughout.

Re-segmentation. Identical byte-pair merges (Sennrich, Haddow and Birch 2016) applied to all corpora leave the controls' ratio h_2/h_1 within 0.72–0.85 at every inventory size, whereas the Voynich ratio rises only from 0.54 to 0.74 after 80 merges (Table S26). The low conditional entropy is not an artefact of a letter being split into several glyphs.

S2. Exploratory profile: tables

Basic statistics and the five properties.

Table S1. Basic token- and glyph-level statistics. Reference corpora are contiguous chunks matched to the Voynich token count (Hebrew: whole text). h_1 = unigram entropy, h_2 = conditional entropy given the previous symbol (bits per symbol, space included). Zipf slope: log-log slope over the 200 most frequent types.

Corpus	Tokens	Types	TTR	Hapax	Mean length	h_1	h_2	h_2/h_1	Zipf slope
Voynich, paragraph text	34,886	6,875	0.197	0.67	4.96	3.87	2.10	0.54	−0.78
Currier A	11,246	3,140	0.279	0.68	4.75	3.83	2.14	0.56	−0.79
Currier B	23,091	4,724	0.205	0.68	5.07	3.87	1.99	0.51	−0.81
Latin (Caesar)	34,886	8,426	0.242	0.59	6.10	4.02	3.30	0.82	−0.75
German (Faust)	30,959	6,221	0.201	0.63	4.78	4.13	3.19	0.77	−0.95
Italian (Dante)	34,886	6,574	0.188	0.62	3.92	4.01	3.15	0.79	−0.94
English (War and Peace)	34,886	4,416	0.127	0.50	4.43	4.12	3.33	0.81	−0.95
Hebrew (Genesis, consonantal)	17,789	6,319	0.355	0.65	4.41	4.13	3.53	0.85	−0.65

Table S2. Excess mutual information (bits) between adjacent tokens inside lines, for different projections of the token pair, relative to a within-line permutation null with identical marginals. Line-initial and line-final tokens are excluded. Reference corpora are cut into the Voynich line-length distribution.

Corpus	token → token	last glyph → first glyph	last two → first two	last glyph → token	token → first glyph
Voynich (all)	0.076	0.146	0.184	0.138	0.142
Currier A	0.047	0.082	0.109	0.076	0.077
Currier B	0.081	0.187	0.227	0.159	0.172

all, uncertain spaces as spaces	0.127	0.182	0.235	0.174	0.191
Latin	0.242	0.037	0.212	0.234	0.167
Italian	0.409	0.152	0.422	0.337	0.356
German	0.216	0.056	0.173	0.191	0.156
Hebrew	0.178	0.047	0.332	0.212	0.218

Table S3. Excess mutual information (bits) for the pair (last token of line k , first token of line $k+1$) within paragraphs, compared with adjacent within-line pairs subsampled to the same n (mean $\pm$ s.d. over 8 subsamples). Reference corpora were cut into lines and paragraphs with the Voynich length distributions.

Corpus	n pairs	across line break	adjacent, same n	adjacent inner, same n	across, distance 3	within, distance 3
Voynich (tokens as transcribed)	3,820	0.005	0.062 $\pm$ 0.007	0.071 $\pm$ 0.006	0.025	0.019 $\pm$ 0.005
variant glyphs collapsed	3,820	0.022	0.145 $\pm$ 0.015	0.162 $\pm$ 0.012	0.051	0.051 $\pm$ 0.009
Currier A, collapsed	1,408	0.019	0.052 $\pm$ 0.016	0.058 $\pm$ 0.015	0.013	0.010 $\pm$ 0.013
Currier B, collapsed	2,348	0.017	0.145 $\pm$ 0.020	0.165 $\pm$ 0.010	0.016	0.039 $\pm$ 0.015
Latin	3,488	0.077	0.070 $\pm$ 0.008	0.067 $\pm$ 0.003	0.008	0.007 $\pm$ 0.004
Italian	3,488	0.166	0.171 $\pm$ 0.012	0.166 $\pm$ 0.018	0.000	0.009 $\pm$ 0.004
German	3,079	0.055	0.060 $\pm$ 0.010	0.062 $\pm$ 0.009	0.012	0.001 $\pm$ 0.006

Table S4. Predictability of token boundaries from k glyphs on each side (model fitted on half the lines, evaluated on the other half). Gain = $1 - H(\text{boundary} | \text{context})/H(\text{boundary})$. F_1 for predicting a boundary at threshold 0.5.

Corpus	boundary rate	gain, $k = 1$	F_1 , $k = 1$	gain, $k = 2$	F_1 , $k = 2$	gain, $k = 3$
Voynich	0.167	73.7%	0.85	77.2%	0.89	60.2%
Currier A	0.173	–	–	70.4%	0.86	–
Currier B	0.166	–	–	79.5%	0.89	–
Latin	0.147	38.7%	0.55	61.6%	0.79	35.0%
Italian	0.232	43.1%	0.63	53.6%	0.76	19.7%
German	0.189	44.9%	0.66	57.8%	0.77	30.4%
Hebrew	0.206	40.1%	0.57	42.3%	0.65	17.8%
Latin, spaces removed and re-inserted by a glyph rule	0.145	73.2%	0.84	73.3%	0.87	43.2%

Table S5. Lexicon test. A within-token glyph model with two glyphs of context is fitted on a random half of the tokens; the share of held-out real tokens (types) whose string occurs in the training half is compared with the same share for tokens sampled from the model.

Corpus	real tokens attested	synthetic tokens attested	real types attested	synthetic types attested	bits/glyph real	bits/glyph synthetic
Voynich	0.79	0.76	0.36	0.30	1.90	1.89

Currier A	0.71	0.70	0.34	0.32	1.95	1.93
Currier B	0.79	0.75	0.36	0.31	1.80	1.79
Latin	0.81	0.20	0.48	0.03	2.68	2.67
German	0.83	0.34	0.44	0.07	2.61	2.55
Italian	0.84	0.52	0.44	0.09	2.75	2.70
Hebrew	0.68	0.36	0.40	0.14	3.00	2.81
Latin, rule-spaced	0.62	0.28	0.28	0.09	2.74	2.76
Latin, syllable pairs	0.65	0.09	0.42	0.09	2.99	2.94
Latin, 3-letter chunks	0.96	0.26	0.82	0.25	3.15	3.07

Table S6. Conditional entropy $h(k) = H(x \mid k \text{ previous symbols})$ of the glyph stream (space as a symbol, lines as separate strings), and differences to a synthetic twin sampled from the order-3 model fitted to the same corpus (same number and length of lines). lzma: compressed bits per symbol.

Corpus	h(1)	h(2)	h(3)	h(4)	h(5)	h(3) – twin	h(4) – twin	h(5) – twin	lzma	lzma twin
Voynich	2.18	1.91	1.82	1.72	1.57	−0.03	−0.06	−0.06	2.40	2.58
Currier A	2.23	1.95	1.83	1.68	1.46	−0.01	−0.03	−0.03	2.62	2.76
Currier B	2.06	1.78	1.70	1.60	1.46	−0.02	−0.04	−0.05	2.33	2.46
Latin	3.38	2.72	2.11	1.66	1.33	0.00	−0.30	−0.30	2.58	3.11
Italian	3.27	2.77	2.31	1.88	1.43	+0.02	−0.19	−0.18	3.04	3.38
German	3.29	2.63	2.10	1.71	1.38	+0.01	−0.19	−0.17	2.88	3.22
Hebrew	3.63	2.90	2.22	1.60	1.05	+0.09	−0.13	−0.15	3.41	3.83
Latin, rule-spaced	3.21	2.75	2.25	1.83	1.39	+0.01	−0.24	−0.30	2.64	3.21

Mechanism battery and robustness.

Table S7. Mechanism battery. All generators produce text in exactly the line and token shape of the Voynich paragraph text. Edge MI and token MI: excess over within-line permutation; line-break MI: (last, first) pair across line breaks; residual: lzma bits per glyph of the text minus that of an order-3 section-drift twin fitted to the same text (negative = redundancy beyond the automaton). The page-drift automaton is fitted in-sample (see text).

Mechanism	edge MI	token MI	line-break MI	space gain	hapax	Zipf	attested real	attested synth.	h(2)	h(4)	residual	I(token; page)
Voynich (ZL3b)	0.146	0.078	0.005	73.7%	0.71	−0.75	0.79	0.75	1.91	1.72	−0.039	0.232
Glyph automaton, order 3, section drift	0.167	0.050	0.005	73.8%	0.71	−0.77	0.79	0.76	1.92	1.75	+0.033	0.134
Glyph automaton, order 4, section drift	0.149	0.069	−0.001	74.1%	0.64	−0.74	0.81	0.76	1.92	1.69	−0.030	0.138
Glyph automaton, order 4, page drift ($\lambda = 0.5$)	0.158	0.076	0.004	73.8%	0.63	−0.74	0.81	0.75	1.92	1.70	−0.052	0.215
Latin, running text	0.034	0.245	0.076	38.9%	0.59	−0.76	0.80	0.20	2.72	1.65	−0.397	0.062

order 4, λ = 0.7	0.164	0.065	-0.002	74.3%	0.64	-0.74	0.81	0.75	1.92	1.69	-0.084	0.279
order 5, λ = 0.3	0.162	0.104	-0.001	73.9%	0.55	-0.73	0.83	0.74	1.92	1.70	-0.110	0.163

Table S9. Naibbe cipher variants: one card per letter or per token; with and without the 3% random space removal; independent sentences per line or running text; Latin or Italian plaintext.

Variant	edge MI	token MI	line- break MI	space gain	hapax	Zipf	attested real	attested synth.	h(2)	h(4)	residual	I(token; page)
Latin, independent lines, card per letter	0.003	0.055	0.002	77.4%	0.49	-1.00	0.85	0.72	1.90	1.70	-0.065	0.007
Latin, independent lines, card per token	0.003	0.120	0.004	77.5%	0.44	-0.75	0.95	0.53	1.88	1.60	-0.198	-0.001
Latin, independent lines, no space removal	0.002	0.069	0.009	80.8%	0.41	-1.00	0.88	0.73	1.87	1.68	-0.074	-0.010
Latin, running text	0.002	0.078	0.023	77.0%	0.49	-1.01	0.84	0.72	1.91	1.71	-0.050	-0.007
Italian, running text	0.002	0.066	0.029	76.8%	0.47	-1.01	0.86	0.69	1.92	1.73	-0.065	0.001

Table S10. Robustness to transcription and to the treatment of uncertain spaces.

Transcription	tokens	types	edge MI	token MI	line- break MI	space gain	hapax	attested real / synth.	h(2)	h(4)	residual	I(token; page)
Zandbergen- Landini, uncertain spaces joined (baseline)	32,399	7,832	0.146	0.078	0.005	73.7%	0.71	0.79 / 0.75	1.91	1.72	-0.039	0.232
Zandbergen- Landini, uncertain spaces as spaces	34,859	6,863	0.182	0.128	0.004	74.0%	0.67	0.83 / 0.81	1.89	1.71	-0.044	0.243
Takahashi	34,384	7,054	0.165	0.116	0.011	75.2%	0.68	0.83 / 0.79	1.89	1.71	-0.039	0.245

Table S11. Out-of-sample comparison (automaton fitted per section on the odd-numbered pages, evaluated on the even-numbered pages) and the battery restricted to single scribal hands and to the two Currier languages.

Subset	edge MI	token MI	line- break MI	space gain	hapax	Zipf	attested real / synth.	h(2)	h(4)	I(token; page)
Held-out real pages (even)	0.150	0.034	-0.001	74.1%	0.71	-0.74	0.75 / 0.73	1.89	1.66	0.289
Twin sampled from the odd-page model	0.150	0.007	0.000	73.8%	0.71	-0.77	0.74 / 0.73	1.92	1.70	0.168
Hand 1 (112 pages)	0.079	0.060	0.001	70.5%	0.71	-0.77	0.71 / 0.71	1.93	1.65	0.129
Hand 2 (46 pages)	0.195	0.087	-0.001	77.4%	0.70	-0.84	0.78 / 0.74	1.72	1.50	0.237
Hand 3 (31 pages)	0.185	0.041	0.006	74.8%	0.72	-0.77	0.71 / 0.69	1.79	1.56	0.182
Currier A	0.082	0.049	0.007	70.6%	0.72	-0.76	0.71 / 0.72	1.95	1.68	0.131
Currier B	0.187	0.081	0.007	76.8%	0.71	-0.78	0.78 / 0.76	1.78	1.60	0.224

Table S12. Leave-one-quire-out jackknife standard errors for the core Voynich statistics (16 quires).

Statistic	estimate	jackknife s.e.	95% interval
edge MI	0.146	0.024	0.099 to 0.193
token MI	0.081	0.024	0.034 to 0.127
line-break MI	0.004	0.010	− 0.016 to 0.024
adjacent MI at the line-break n	0.058	0.022	0.015 to 0.101
space gain (k = 1)	0.737	0.018	0.702 to 0.772
hapax share	0.710	0.006	0.698 to 0.722
Zipf slope	− 0.753	0.034	− 0.819 to − 0.687
attested, real	0.792	0.026	0.742 to 0.843
attested, synthetic	0.754	0.025	0.704 to 0.803
lexicon gap (real − synthetic)	0.039	0.021	− 0.003 to 0.080
h(2)	1.905	0.047	1.812 to 1.998
h(4)	1.717	0.025	1.669 to 1.765

Drift, hand, illustrations and labels.

Table S13. Mean Jensen–Shannon divergence between the glyph-bigram distributions of two pages (450-glyph samples) as a function of their folio distance, within the same section and Currier language.

Pages	d = 1–2	3–6	7–14	15–40	> 40	other section
Voynich, Currier A	0.124	0.145	0.150	0.150	0.161	0.193
Voynich, Currier B	0.126	0.137	0.134	0.147	0.147	0.169
Section-drift twin, A	0.093	0.093	0.094	0.093	0.094	0.148
Section-drift twin, B	0.086	0.085	0.085	0.092	0.092	0.122
Latin in book order (chunked like A)	0.183	0.188	0.195	0.192	0.192	0.193

Table S14. Mean Jensen–Shannon divergence between pages grouped by scribal hand (palaeographic assignment of Davis 2020 as recorded in the ZL metadata) and illustration section.

Pairs	same section, same hand	same section, different hand	different section, same hand	different section, different hand
Currier B, all sections	0.135	0.153	0.172	0.168
Herbal-B, folio distance ≤ 6 pages	0.138	0.182	–	–
Herbal-B, folio distance > 20 pages	0.157	0.165	–	–

Table S15. Mantel tests between text distance and illustration-feature distance across herbal pages (pairs closer than 3 folios excluded).

Page set	text representation	Mantel r	p (permutation)	text vs folio distance r (p)
Herbal-A, hand 1 (85 pages)	token frequencies	− 0.055	0.79	+ 0.115 (0.001)
	glyph-bigram frequencies	− 0.070	0.87	–
Herbal-B, hands 2 and 5 (25 pages), unstratified	token frequencies	+ 0.223	0.046	+ 0.163 (0.044)
permutation within hand	token frequencies	+ 0.223	0.076	–

Section	pages	label tokens	label types	share recurring on another page	null: text tokens of the same pages
Zodiac	12	330	271	0.276	0.363
Astronomical	5	142	136	0.028	0.273
Pharmaceutical	15	234	219	0.090	0.364
Biological	11	122	111	0.115	0.342
Cosmological	5	196	181	0.020	0.277

Neighbour similarity and line structure. Adjacent inner tokens share their first or last glyphs, or are identical, as often as tokens of the same line in random order (ratios 0.97 to 1.03), and single-glyph edits between neighbours are not directional (largest $z = 1.8$).

Table S20. Share of adjacent inner tokens that share the given feature, observed / within-line permutation null = ratio.

Corpus	n pairs	first glyph	first two	last glyph	last two	last three	identical
Voynich	19,895	0.193 / 0.197 = 0.98	0.120 / 0.124 = 0.97	0.311 / 0.314 = 0.99	0.168 / 0.163 = 1.03	0.093 / 0.091 = 1.02	0.011 / 0.011 = 0.98
Currier A	5,661	0.210 / 0.212 = 0.99	0.125 / 0.125 = 1.00	0.253 / 0.251 = 1.01	0.132 / 0.130 = 1.02	0.062 / 0.061 = 1.01	0.012 / 0.012 = 0.94
Currier B	13,980	0.186 / 0.191 = 0.98	0.118 / 0.124 = 0.95	0.335 / 0.339 = 0.99	0.183 / 0.178 = 1.03	0.106 / 0.103 = 1.03	0.010 / 0.011 = 0.94
Latin	22,441	0.055 / 0.063 = 0.87	0.010 / 0.016 = 0.62	0.150 / 0.135 = 1.10	0.044 / 0.041 = 1.07	0.008 / 0.012 = 0.68	0.000 / 0.002 = 0.00
Italian	22,441	0.046 / 0.061 = 0.76	0.006 / 0.014 = 0.45	0.185 / 0.159 = 1.16	0.013 / 0.017 = 0.76	0.001 / 0.006 = 0.20	0.000 / 0.005 = 0.09
German	19,859	0.051 / 0.065 = 0.79	0.011 / 0.018 = 0.60	0.120 / 0.124 = 0.97	0.040 / 0.046 = 0.86	0.008 / 0.016 = 0.48	0.002 / 0.008 = 0.24
Hebrew	11,373	0.105 / 0.129 = 0.81	0.025 / 0.035 = 0.73	0.114 / 0.119 = 0.95	0.022 / 0.029 = 0.74	0.007 / 0.014 = 0.48	0.001 / 0.006 = 0.24

Table S21. Single-glyph edits between adjacent inner tokens at edit distance 1 (739 pairs), forward versus inverse counts; $z = (n_{ab} - n_{ba})/\sqrt{(n_{ab} + n_{ba})}$. Share of pairs in edits whose inverse is less than half as frequent: 0.18.

edit a → b	count a → b	count b → a	z
ins t @mid	14	6	1.8
sub r>l @end	36	23	1.7
del d @mid	30	20	1.4
ins i @mid	16	10	1.2
del e @mid	50	40	1.1
sub c>s @start	17	13	0.7
del o @start	18	14	0.7
sub t>k @mid	27	24	0.4
ins q @start	37	35	0.2

Table S22. Similarity of a token to candidate copy sources (Timm and Schinner 2020) and to controls.

Relation	pairs	mean edit distance	P(identical)	P(ED ≤ 1)	P(ED ≤ 2)
previous token in the line	30,756	4.44	0.009	0.046	0.126
token above (same index)	29,512	4.50	0.010	0.044	0.125

S5. Held-out statistics for every generator

Table S30. E2: held-out statistics (test: even bifolios).

Statistic	Voynich	full	no recency	no pruning	no session	plain automaton
hapax share †	0.7085	0.6022 ✗	0.6314 ✗	0.6799 ✗	0.6275 ✗	0.7044 ✓
type-token ratio †	0.3012	0.2190 ✗	0.2110 ✗	0.2825 ✓	0.2261 ✗	0.2914 ✓
Zipf slope	−0.7460	−0.7448 ✓	−0.7588 ✓	−0.7537 ✓	−0.7460 ✓	−0.7633 ✓
mean word length	5.3867	5.1717 ✓	5.1360 ✓	5.2468 ✓	5.1550 ✓	5.2544 ✓
boundary MI	0.1523	0.1328 ✓	0.1645 ✓	0.1315 ✓	0.1325 ✓	0.1563 ✓
word-pair MI	0.0549	0.0255 ✓	0.0582 ✓	0.0122 ✓	0.0304 ✓	0.0340 ✓
distance-2 word MI	0.0425	0.0233 ✓	0.0192 ✓	0.0103 ✗	0.0095 ✗	0.0153 ✗
line-break word MI	0.0006	−0.0014 ✓	0.0038 ✓	0.0010 ✓	0.0040 ✓	0.0021 ✓
line-break boundary MI	0.0120	−0.0032 ✓	0.0163 ✓	0.0052 ✓	0.0016 ✓	−0.0009 ✓
boundary MI, equal n	0.1511	0.1370 ✓	0.1660 ✓	0.1246 ✓	0.1272 ✓	0.1406 ✓
space predictability	0.7388	0.7847 ✗	0.7910 ✗	0.7464 ✓	0.7772 ✗	0.7375 ✓
lexicon test, real words	0.7378	0.8133 ✗	0.8255 ✗	0.7487 ✓	0.8089 ✗	0.7481 ✓
lexicon test, synthetic words	0.7186	0.7657 ✓	0.7779 ✓	0.7183 ✓	0.7613 ✓	0.7127 ✓
conditional entropy h2	1.8787	1.8107 ✓	1.7747 ✗	1.9186 ✓	1.8226 ✓	1.9219 ✓
conditional entropy h3	1.7779	1.7097 ✓	1.6695 ✗	1.7958 ✓	1.7130 ✓	1.7954 ✓
conditional entropy h4	1.6475	1.6415 ✓	1.6066 ✓	1.6929 ✓	1.6385 ✓	1.6909 ✓
lzma bits per character	2.4440	2.3285 ✓	2.2877 ✓	2.4836 ✓	2.3491 ✓	2.4943 ✓
page MI †	0.2529	0.3048 ✓	0.2172 ✓	0.1993 ✓	0.2168 ✓	0.1731 ✗
T1 line-break ratio	0.0809	−0.1428 ✓	−0.0347 ✓	0.2131 ✓	−0.0763 ✓	−0.0125 ✓
T2a final form, margin	0.0942	0.1346 ✓	0.1441 ✓	0.1230 ✓	0.1268 ✓	0.1357 ✓
T2a final form, paragraph end	0.0373	0.1653 ✗	0.1770 ✗	0.1653 ✗	0.1691 ✗	0.1614 ✗
T2b line-fill slack	1.9622	9.4848 ✗	9.3461 ✗	10.5469 ✗	9.6503 ✗	10.3487 ✗
T2c dittography	0.6098	1.9992 ✓	1.3019 ✓	1.6935 ✓	1.4505 ✓	0.9910 ✓
T3a zero replication †	0.0384	0.0403 ✓	0.0391 ✓	0.0518 ✗	0.0473 ✗	0.0551 ✗
T3b slot consistency	0.8683	0.8777 ✓	0.8835 ✓	0.8604 ✓	0.8768 ✓	0.8576 ✓
T3b monotone share	0.5471	0.5828 ✓	0.5943 ✓	0.5541 ✓	0.5804 ✓	0.5532 ✓
T3c final-position entropy	0.6299	0.6535 ✓	0.6534 ✓	0.6593 ✓	0.6540 ✓	0.6665 ✓
T3c initial-position entropy	0.8734	0.8734 ✓	0.8672 ✓	0.8786 ✓	0.8725 ✓	0.8806 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.7808	0.8672 ✓	0.6696 ✓	0.6003 ✓	0.5872 ✓	0.5068 ✓
T6a recency, near repeats †	0.1399	0.1451 ✓	−0.0081 ✗	0.1058 ✓	0.0923 ✗	−0.0130 ✗
T6a recency, exact repeats †	0.2324	0.1925 ✓	0.0050 ✗	0.2737 ✓	0.3201 ✓	−0.0224 ✓
T6b within-page drift †	−0.0082	−0.0016 ✗	−0.0010 ✗	−0.0015 ✗	−0.0017 ✗	−0.0006 ✗
T7b deviant rate †	26.8523	31.6494 ✓	35.6074 ✓	47.0364 ✓	37.9130 ✓	44.4175 ✓
T7b deviant recurrence †	0.0047	0.1868 ✗	0.1458 ✗	0.1214 ✗	0.1520 ✗	0.0906 ✗
T7b slip share	0.2870	0.3955 ✓	0.4417 ✗	0.2873 ✓	0.4148 ✓	0.2741 ✓
E1 line step	0.0591	0.0024 ✓	0.0419 ✓	0.0091 ✓	−0.0004 ✓	0.0070 ✓
E1 paragraph step	−0.0549	−0.0440 ✓	−0.0018 ✓	−0.0164 ✓	−0.0035 ✓	−0.0103 ✓
E1 within-line decay	0.0608	0.0002 ✓	0.0061 ✓	0.0036 ✓	−0.0168 ✓	−0.0154 ✓
E1 within-paragraph decay	0.0447	0.0445 ✓	−0.0080 ✓	0.0290 ✓	0.0036 ✓	−0.0448 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

Table S31. E3 primary direction (fit even, test odd).

Statistic	Voynich	full	no margin rule	no drift	no slips etc.	E2 class
hapax share †	0.7199	0.7227 ✓	0.6800 ✗	0.7173 ✓	0.6246 ✗	0.5970 ✗
type-token ratio †	0.2871	0.3223 ✓	0.2323 ✓	0.3145 ✓	0.2065 ✗	0.1871 ✗
Zipf slope	-0.7499	-0.7776 ✓	-0.7831 ✓	-0.7783 ✓	-0.7772 ✓	-0.7929 ✓
mean word length	5.2924	5.4708 ✓	5.2914 ✓	5.4364 ✓	5.3135 ✓	5.2861 ✓
boundary MI	0.1388	0.1264 ✓	0.1428 ✓	0.1369 ✓	0.1498 ✓	0.1343 ✓
word-pair MI	0.0594	-0.0388 ✗	0.0345 ✓	-0.0615 ✗	-0.0024 ✓	0.0452 ✓
distance-2 word MI	0.0464	0.0168 ✓	0.0260 ✓	0.0161 ✓	0.0392 ✓	0.0413 ✓
line-break word MI	0.0002	0.0047 ✓	0.0031 ✓	0.0027 ✓	0.0173 ✓	0.0042 ✓
line-break boundary MI	0.0078	0.0059 ✓	0.0114 ✓	0.0057 ✓	0.0061 ✓	0.0099 ✓
boundary MI, equal n	0.1480	0.1405 ✓	0.1341 ✓	0.1483 ✓	0.1407 ✓	0.1366 ✓
space predictability	0.7343	0.6962 ✓	0.7540 ✓	0.7009 ✓	0.7816 ✗	0.7896 ✗
lexicon test, real words	0.7455	0.7103 ✓	0.7992 ✓	0.7175 ✓	0.8277 ✓	0.8444 ✗
lexicon test, synthetic words	0.7251	0.6753 ✗	0.7492 ✓	0.6759 ✗	0.7617 ✓	0.7762 ✓
conditional entropy h2	1.9100	1.9981 ✓	1.8345 ✓	1.9888 ✓	1.7647 ✗	1.7471 ✗
conditional entropy h3	1.8075	1.8477 ✓	1.7161 ✓	1.8397 ✓	1.6595 ✗	1.6535 ✗
conditional entropy h4	1.6769	1.7241 ✓	1.6352 ✓	1.7190 ✓	1.6023 ✗	1.5997 ✗
lzma bits per character	2.4581	2.5279 ✓	2.3318 ✓	2.5210 ✓	2.2590 ✓	2.2254 ✗
page MI †	0.3195	0.2646 ✓	0.2450 ✓	0.2530 ✓	0.3493 ✓	0.3620 ✓
T1 line-break ratio	0.3137	-0.1355 ✓	0.0043 ✓	-0.1009 ✓	0.0489 ✓	-0.0051 ✓
T2a final form, margin	0.0789	0.0464 ✓	0.0212 ✗	0.0480 ✓	0.0566 ✓	0.0180 ✗
T2a final form, paragraph end	0.0428	0.0538 ✓	0.0167 ✗	0.0578 ✓	0.0587 ✓	0.0206 ✓
T2b line-fill slack	2.2694	2.6287 ✗	8.0667 ✗	2.6204 ✗	2.7059 ✗	8.1219 ✗
T2c dittography	0.9501	1.3470 ✓	1.6969 ✓	2.1210 ✗	1.3516 ✓	1.7792 ✓
T3a zero replication †	0.0415	0.0574 ✗	0.0547 ✗	0.0621 ✗	0.0313 ✗	0.0298 ✗
T3b slot consistency	0.8684	0.8540 ✓	0.8745 ✓	0.8542 ✓	0.8828 ✓	0.8848 ✓
T3b monotone share	0.5607	0.5277 ✓	0.5785 ✓	0.5353 ✓	0.5936 ✓	0.6047 ✓
T3c final-position entropy	0.6217	0.6402 ✓	0.6147 ✓	0.6358 ✓	0.5886 ✓	0.5839 ✓
T3c initial-position entropy	0.8812	0.8772 ✓	0.8692 ✓	0.8809 ✓	0.8638 ✓	0.8631 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.8021	0.8980 ✓	0.6265 ✓	0.8263 ✓	0.7591 ✓	0.7922 ✓
T6a recency, near repeats †	0.1635	0.1168 ✓	0.1155 ✓	0.1290 ✓	0.0478 ✗	0.1141 ✓
T6a recency, exact repeats †	0.2042	0.2052 ✓	0.0855 ✓	0.1749 ✓	0.0136 ✗	0.1911 ✓
T6b within-page drift †	-0.0084	-0.0019 ✗	-0.0020 ✗	-0.0012 ✗	-0.0016 ✗	-0.0021 ✗
T7b deviant rate †	23.4977	49.0845 ✗	31.9529 ✓	50.6217 ✗	22.9400 ✓	18.7890 ✓
T7b deviant recurrence †	0.0086	0.0681 ✗	0.0487 ✗	0.0642 ✗	0.1837 ✗	0.1836 ✗
T7b slip share	0.2723	0.2960 ✓	0.4209 ✗	0.2797 ✓	0.4257 ✗	0.4163 ✗
E1 line step	0.0655	0.0034 ✓	0.0199 ✓	0.0189 ✓	0.0096 ✓	0.0123 ✓
E1 paragraph step	-0.0244	-0.0318 ✓	0.0055 ✓	-0.0034 ✓	-0.0054 ✓	-0.0012 ✓
E1 within-line decay	0.0692	0.0165 ✓	0.0066 ✓	0.0020 ✓	0.0075 ✓	-0.0005 ✓
E1 within-paragraph decay	0.0817	0.0353 ✓	0.0435 ✓	0.0234 ✓	0.0009 ✓	0.0298 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

Table S32. E6 primary direction (fit even, test odd).

Statistic	Voynich	two-route	no coinage	no repertoire variation	no recency	no boundary rule
hapax share †	0.7199	0.6070 ✗	0.3152 ✗	0.6781 ✗	0.6526 ✗	0.6047 ✗
type-token ratio †	0.2871	0.1336 ✗	0.0640 ✗	0.2890 ✓	0.1877 ✗	0.1327 ✗
Zipf slope	-0.7499	-0.7634 ✓	-0.7907 ✓	-0.7242 ✓	-0.7807 ✓	-0.7716 ✓
mean word length	5.2924	5.0055 ✗	4.8110 ✗	5.5328 ✗	5.1737 ✓	4.9951 ✗
boundary MI	0.1388	0.1256 ✓	0.1238 ✓	0.1287 ✓	0.1393 ✓	0.0005 ✗
word-pair MI	0.0594	-0.0032 ✓	0.0313 ✓	-0.0468 ✗	-0.0151 ✗	-0.0262 ✗
distance-2 word MI	0.0464	0.0503 ✓	0.0746 ✓	0.0125 ✓	0.0450 ✓	0.0458 ✓
line-break word MI	0.0002	0.0322 ✓	0.0536 ✗	0.0182 ✓	0.0272 ✓	0.0407 ✗
line-break boundary MI	0.0078	0.0139 ✓	0.0189 ✓	0.0180 ✓	0.0189 ✓	0.0195 ✓
boundary MI, equal n	0.1480	0.1365 ✓	0.1387 ✓	0.1434 ✓	0.1449 ✓	0.0041 ✓
space predictability	0.7343	0.8110 ✗	0.8484 ✗	0.7342 ✓	0.7830 ✗	0.8276 ✗
lexicon test, real words	0.7455	0.8978 ✗	0.9677 ✗	0.7491 ✓	0.8471 ✗	0.9015 ✗
lexicon test, synthetic words	0.7251	0.8069 ✗	0.8473 ✗	0.6896 ✓	0.7681 ✓	0.8068 ✗
conditional entropy h2	1.9100	1.7184 ✗	1.6104 ✗	1.9271 ✓	1.7833 ✓	1.7405 ✗
conditional entropy h3	1.8075	1.6332 ✗	1.5380 ✗	1.8163 ✓	1.6911 ✗	1.6564 ✗
conditional entropy h4	1.6769	1.5513 ✗	1.4717 ✗	1.7150 ✓	1.6022 ✓	1.5721 ✗
lzma bits per character	2.4581	2.2290 ✗	2.0911 ✗	2.4963 ✓	2.3219 ✓	2.2539 ✓
page MI †	0.3195	0.4378 ✓	0.4760 ✗	0.1916 ✗	0.4043 ✓	0.3943 ✓
T1 line-break ratio	0.3137	-0.0369 ✓	0.2971 ✓	0.0502 ✓	-0.0398 ✓	undef. ✗
T2a final form, margin	0.0789	0.1025 ✓	0.1033 ✓	0.0932 ✓	0.1019 ✓	0.0996 ✓
T2a final form, paragraph end	0.0428	0.0461 ✓	0.0512 ✓	0.0678 ✓	0.0594 ✓	0.0565 ✓
T2b line-fill slack	2.2694	2.7781 ✗	2.6571 ✗	2.7373 ✗	2.5448 ✗	2.7543 ✗
T2c dittography	0.9501	1.1848 ✓	1.2145 ✓	1.1697 ✓	0.9520 ✓	1.2275 ✓
T3a zero replication †	0.0415	0.0353 ✓	0.0285 ✗	0.0464 ✓	0.0377 ✓	0.0330 ✓
T3b slot consistency	0.8684	0.9086 ✗	0.9284 ✗	0.8616 ✓	0.8913 ✓	0.9071 ✗
T3b monotone share	0.5607	0.6645 ✗	0.7103 ✗	0.5279 ✓	0.6164 ✓	0.6506 ✗
T3c final-position entropy	0.6217	0.6148 ✓	0.6017 ✓	0.6468 ✓	0.6263 ✓	0.6149 ✓
T3c initial-position entropy	0.8812	0.8595 ✓	0.8541 ✓	0.8748 ✓	0.8645 ✓	0.8689 ✓
T4 periodicity	0.0000	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓	0.0000 ✓
T5 drift shape	0.8021	0.4761 ✓	0.3850 ✗	0.5684 ✓	0.5275 ✓	0.4666 ✓
T6a recency, near repeats †	0.1635	0.0641 ✗	0.0368 ✗	0.0134 ✗	-0.0262 ✗	0.0488 ✗
T6a recency, exact repeats †	0.2042	-0.0005 ✗	-0.0052 ✗	0.1216 ✓	-0.0339 ✗	0.0655 ✗
T6b within-page drift †	-0.0084	-0.0002 ✗	-0.0010 ✗	-0.0008 ✗	-0.0006 ✗	-0.0008 ✗
T7b deviant rate †	23.4977	27.3205 ✓	20.4694 ✓	38.1832 ✓	28.4477 ✓	26.9757 ✓
T7b deviant recurrence †	0.0086	0.0467 ✗	0.1503 ✗	0.0942 ✗	0.0602 ✗	0.0499 ✗
T7b slip share	0.2723	0.4817 ✗	0.8110 ✗	0.2929 ✓	0.3348 ✓	0.4602 ✗
E1 line step	0.0655	0.0330 ✓	0.0103 ✓	0.0055 ✓	0.0130 ✓	0.0381 ✓
E1 paragraph step	-0.0244	0.0341 ✓	-0.0033 ✓	0.0250 ✓	-0.0117 ✓	0.0118 ✓
E1 within-line decay	0.0692	0.0018 ✓	0.0074 ✓	-0.0079 ✓	0.0063 ✓	0.0070 ✓
E1 within-paragraph decay	0.0817	0.0209 ✓	0.0033 ✓	0.0348 ✓	-0.0171 ✗	0.0092 ✓

† fitting target on the training half. ✓ within tolerance $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$.

S6. Drift and burstiness

Table S33. E5: page clustering index B by frequency band and gradient G with quire-jackknife 95% intervals.

Corpus	G	B 2–4	B 5–19	B 20–99	B ≥ 100
Voynich	+ 0.24 (− 0.10, 0.59)	2.02	1.83	1.58	1.36
Voynich (Takahashi)	+ 0.19 (− 0.28, 0.66)	1.91	1.87	1.58	1.34
Latin	+ 0.74 (0.57, 0.92)	2.95	2.29	1.40	1.07
Italian	+ 0.78 (0.47, 1.09)	2.59	1.56	1.19	1.02
German	+ 1.06 (0.81, 1.30)	4.89	2.96	1.70	1.13
English	+ 1.08 (0.93, 1.22)	5.11	2.61	1.74	1.22
VB-run	+ 0.66 (0.49, 0.82)	2.23	1.48	1.15	1.03
NAIB-run	+ 0.07 (− 0.33, 0.46)	1.09	1.04	1.02	1.00
LDRIFT	− 0.19 (− 0.52, 0.14)	2.22	3.02	2.67	2.12
REP	− 0.60 (− 1.14, − 0.05)	1.53	2.06	2.77	2.40
HGR2	+ 0.97 (0.49, 1.44)	3.68	1.70	1.40	1.31
MK-sec	+ 0.03 (− 0.28, 0.34)	1.10	1.15	1.07	1.03

Table S34. E4: drift slopes of line similarity over line distance 2–8 with page-bootstrap 95% intervals.

Corpus	L (lexical)	S (sublexical)
Voynich	− 0.0142 (− 0.0275, − 0.0022)	− 0.0052 (− 0.0085, − 0.0023)
Latin	− 0.0115 (− 0.0257, 0.0040)	− 0.0028 (− 0.0045, − 0.0011)
Italian	− 0.0147 (− 0.0255, − 0.0026)	− 0.0121 (− 0.0153, − 0.0088)
German	− 0.0252 (− 0.0402, − 0.0116)	− 0.0043 (− 0.0070, − 0.0015)
English	− 0.0128 (− 0.0206, − 0.0042)	+ 0.0003 (− 0.0031, 0.0040)
VB-run	− 0.0092 (− 0.0190, − 0.0001)	+ 0.0002 (− 0.0020, 0.0022)
NAIB-run	+ 0.0076 (− 0.0075, 0.0242)	− 0.0015 (− 0.0043, 0.0012)
GDRIFT	− 0.0517 (− 0.0660, − 0.0385)	− 0.0148 (− 0.0176, − 0.0112)
LDRIFT	− 0.0398 (− 0.0498, − 0.0291)	+ 0.0007 (− 0.0023, 0.0033)
MK-sec	− 0.0042 (− 0.0151, 0.0064)	− 0.0007 (− 0.0030, 0.0016)

S7. Definitions of the 40 phenotype statistics

Profile statistics (Section 3 of the main text):

- hapax share and type-token ratio;
- Zipf slope of the 200 most frequent types;
- mean word length;
- boundary MI: excess MI between the last glyph of a word and the first glyph of the next inside lines, against a within-line permutation null;
- word-pair MI and distance-2 word MI;
- line-break MI and line-break boundary MI, with the boundary MI at equal n;
- space predictability: information gain about word boundaries from the flanking glyph pair;
- lexicon test: share of held-out real words and of glyph-trigram samples attested in the training vocabulary;

- conditional glyph entropies h2–h4;
- lzma bits per character;
- page MI between page identity and word.

Mechanism battery: T1 page-conditional line-break ratio; T2a final-form excess (margin-bound, paragraph-final); T2b slack ratio (stars section); T2c dittography ratio; T3a zero replication (250-token chunks); T3b slot consistency and monotone share; T3c positional entropy (final, initial); T4 any periodicity peak; T5 step/smooth prediction-error ratio; T6a near and exact recency indices; T6b slope of line similarity over line distance 1–8; T7b deviant rate, recurrence share and slip share.

E1: graded line step, paragraph step, graded within-line decay, within-paragraph decay.

Tolerance: $|G - V| \leq 2\sqrt{SE_V^2 + SD_G^2} + 0.01|V|$. SE_V is a leave-one-quire-out jackknife over the test half; SD_G is taken over three generator seeds. T4 passes if at most one seed shows a peak.

S8. Implementation notes

- **Page-conditional T1 estimator.** Per page, the excess plug-in MI over 200 within-page re-pairings, for the break pairs and for consecutive blocks of within-line pairs of the same size, averaged over pages with weights equal to the number of break pairs. The pooled estimators failed a known-truth simulation in which each page of Latin received a different letter substitution; this estimator returned identical values with and without substitution.
- **E1 expectations.** Position classes: offset from the line start (1 or ≥ 2), offset from the line end (1 or ≥ 2), paragraph-final line or not. The binary expectation is the exact mean over pairs of distinct eligible tokens of the page in the same class pair.
- **E5 null.** For each type, the expected same-page pairs are $\sum_g C(n_{tg}, 2) \cdot \sum_{p \in g} N_p(N_p - 1) / (N_g(N_g - 1))$, with groups $g = \text{hand} \times \text{section}$.
- **E7 coder.** 32-bit Witten–Neal–Cleary arithmetic coding over integer transition counts. The end-of-line suppression below 60% of the target length and the stop at 140% are deterministic and replicated by the receiver.